HSLU Lucerne University of Applied Sciences and Arts

# Lewis structures

CHEM: Chemistry

Dr. Rebecca Ravotti
HS25

**Engineering and Architecture**
October 13, 2026

FH Zentralschweiz

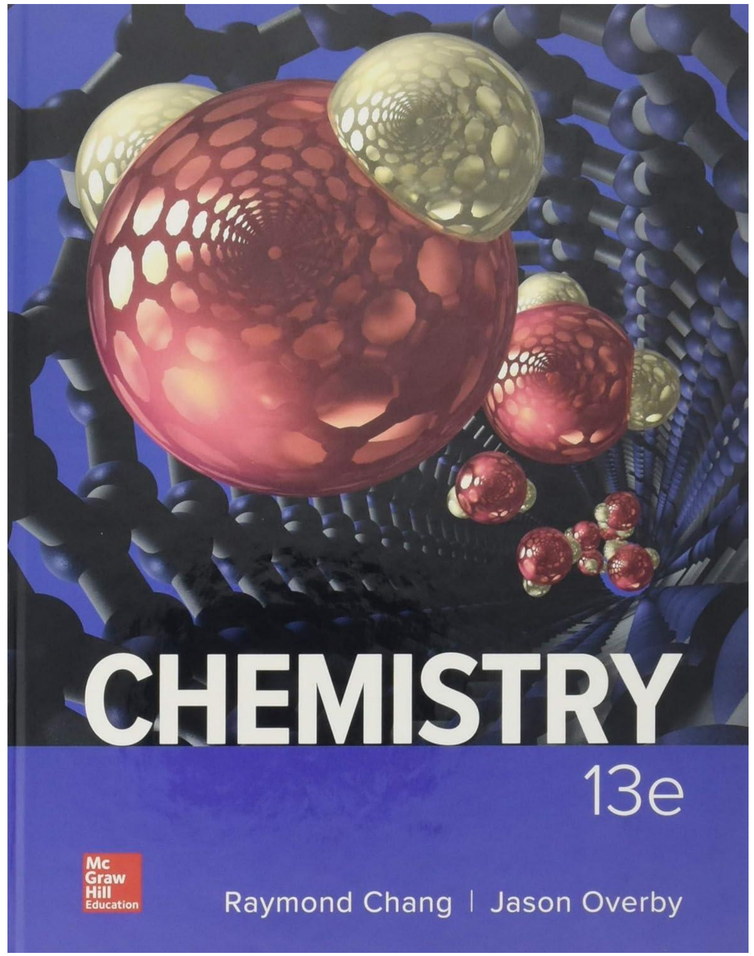
# Course Details

| Lecturer | Date | Week | Topic |
|---|---|---|---|
| Macarena San Martin | 16.09.2026 | 1 | Introduction to Matter |
| Macarena San Martin | 23.09.2026 | 2 | Atomic Structure and Radioactivity |
| Macarena San Martin | 30.09.2026 | 3 | Atomic Models (Self Study), Periodic table and electronic configuration |
| Rebecca Ravotti | 07.10.2026 | 4 | Chemical Bonds and Hybridization |
| Rebecca Ravotti | 14.10.2026 | 5 | Lewis Structure |
| Davide Toniolo | 21.10.2026 | 6 | Chemical Reactions |
| Davide Toniolo | 28.10.2026 | 7 | Acids & Bases |
| Davide Toniolo | 04.11.2026 | 8 | Acids & Bases Exercises |
| Davide Toniolo | 11.11.2026 | 9 | Redox Reactions |
| Davide Toniolo | 18.11.2026 | 10 | Redox exercises |
| Rebecca Ravotti | 25.11.2026 | 11 | Metals and Semimetals |
| Rebecca Ravotti | 02.12.2026 | 12 | Organic Compounds nomenclature |
| Rebecca Ravotti | 09.12.2026 | 13 | Organic Exercises |
| Macarena /Rebecca/Davide | 16.12.2026 | 14 | Final Laboratory |

# Recommended literature

Chemistry ; Authors, Raymond Chang, Jason Overby ; Edition, **13, illustrated** ; Publisher, McGraw-Hill Education, 2018 ; ISBN, 1260085317, 9781260085310

Periodic table: **https://ptable.com**



# Structure of the Lecture

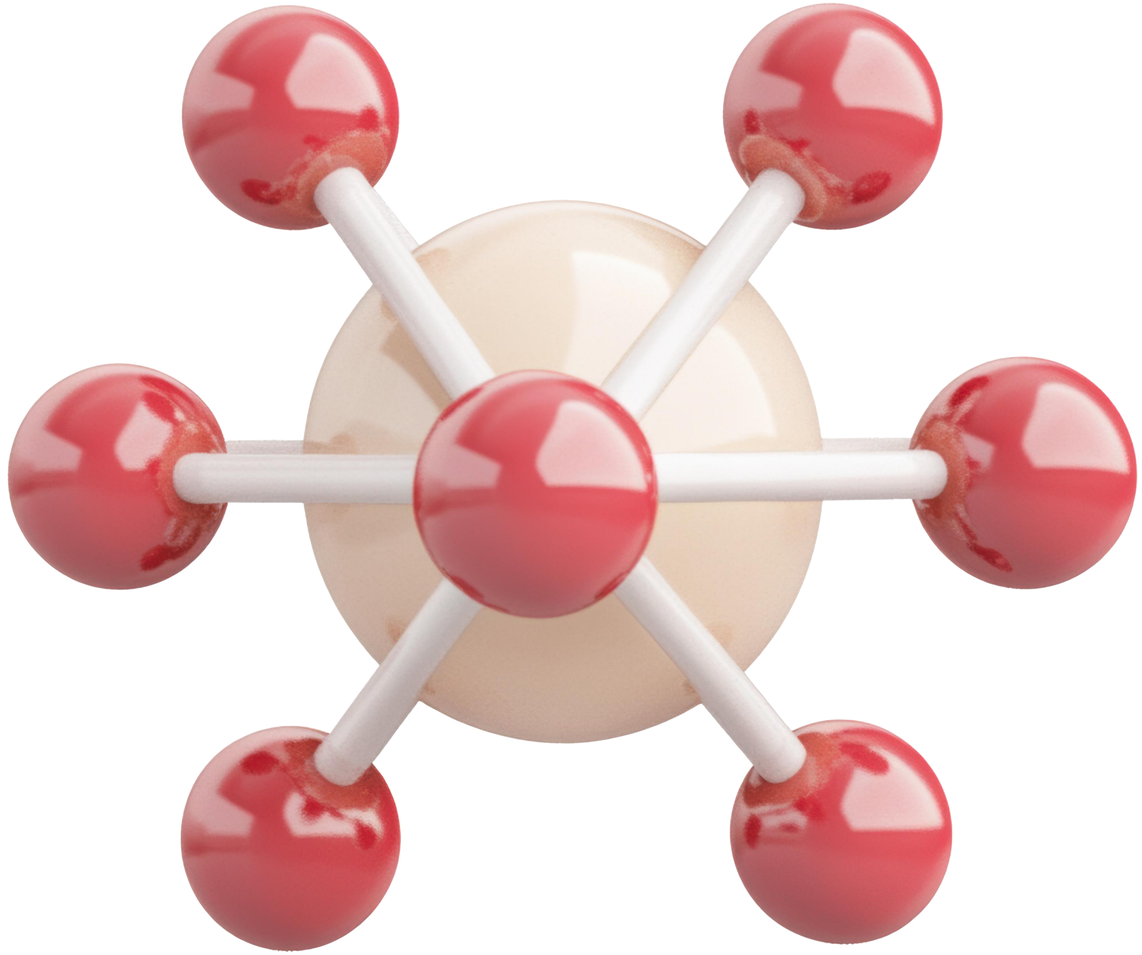
- Introduction/learning objectives (10 min) (08:30-08:40)
- Recap bonds + open QA (20 min) (08:40 – 09:00)
- Kahoot (10 min) (09:00 – 09:10)

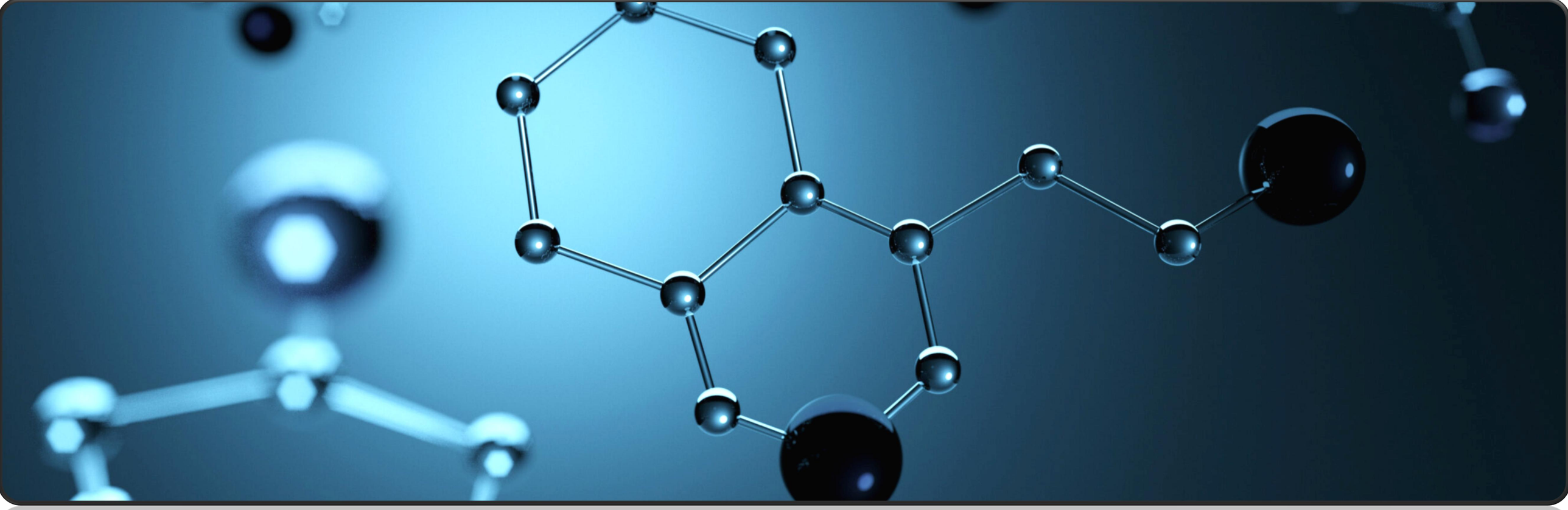
Break (20 min) (09:10 – 09:25)

- Theoretical input (15 min) (09:25 – 09:40)
- Exercise guided (20 min) (09:40 – 10:00)
- Exercise independent (10 min) (10:00 – 10:10)

Break (20 min) (10:10 – 10:30)

- Exercises (15 min) (10:30 – 10:45)
- Wrap-up/Feedback (15 min) (10:40 – 10:55)





# Today's Content

- Recap hybridisation
- Recap bonds
- Lewis structures – how to draw bonds
- VSEPR – chemical geometry
- Exercises

# Learning objectives

Recap hybridisation

Recap bonds

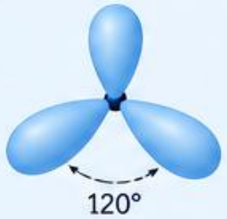
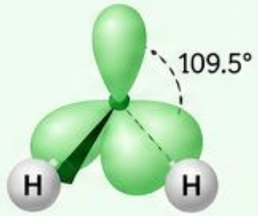
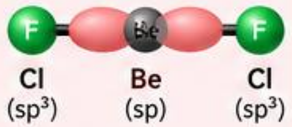
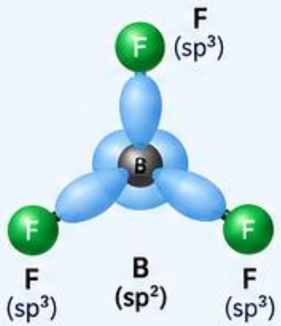
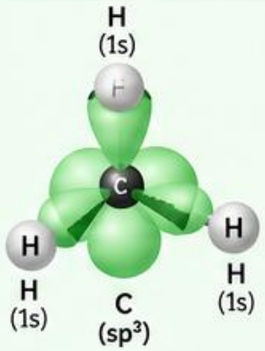
Get to know Lewis formulas

**Be able to draw chemical bonds**

Be able to assign charges

# Recap hybridisation

- **Hybridisation** = mixing of atomic orbitals (s and p) on the same atom to form new, equivalent hybrid orbitals **suitable for bonding.**
- Occurs within one atom (before bonding).
- Leaves 0, 1 or 2 unhybridised p orbitals (which can form π bonds).

| | **sp**<br>1 s + 1 p → 2 sp | **$sp^2$**<br>1 s + 2 p → 3 $sp^2$ | **$sp^3$**<br>1 s + 3 p → 4 $sp^3$ |
|---|---|---|---|
| Number of hybrid orbitals | 2 | 3 | 4 |
| Unhybridised p orbitals left | 2 | 1 | 0 |
| Geometry | Linear | Trigonal planar | Tetrahedral |
| Bond angle | 180° | 120° | 109.5° |
| Orbital picture (hybrid orbitals) |  |  |  |
| Example molecule | **$BeCl_2$** | **$BF_3$** | **$CH_4$** |
| Structure (with orbitals) |  |  |  |
| | Two sp orbitals form σ bonds with Cl. Two unhybridised p orbitals remain (perpendicular). | Three $sp^2$ orbitals form σ bonds with F. One unhybridised p orbital (perpendicular to the plane). | Four $sp^3$ orbitals form σ bonds with H. No unhybridised p orbitals left. |

## Covalent bonds and orbital overlap

### σ bond (sigma)

- Head-on overlap of orbitals
- Cylindrical electron density along the internuclear axis
- **Stronger bond**



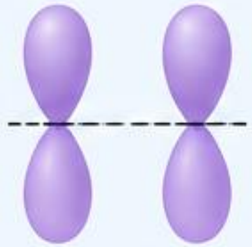
**s–s σ bond**
(e.g. H–H)



**$sp^3$–$sp^3$ σ bond**
(e.g. C–C in ethane)

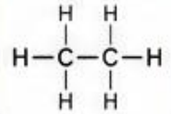
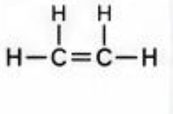
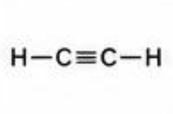
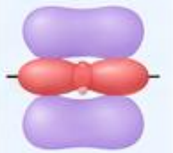
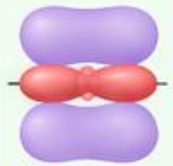
### π bond (pi)

- Sideways overlap of unhybridised p orbitals
- Electron density above and below the internuclear axis
- Weaker bond than σ bond

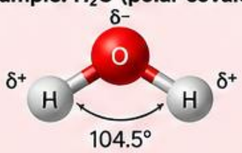
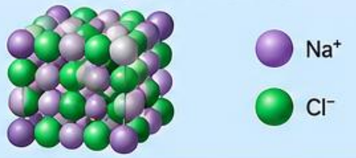
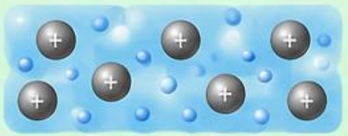
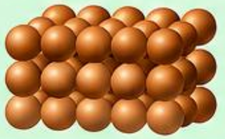
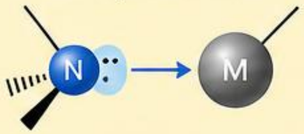
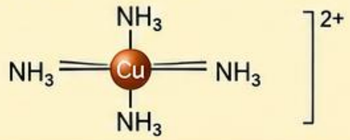


**p–p π bond**
**(e.g. C=C)**

### Single, double and triple bonds

| Single bond | Double bond | Triple bond |
|---|---|---|
|  |  |  |
| 1 σ bond<br>(0 π bonds) | 1 σ + 1 π bond | 1 σ + 2 π bonds |
|  |  |  |
| e.g. ethane<br>($C_2H_6$) | e.g. ethene<br>($C_2H_4$) | e.g. ethyne<br>($C_2H_2$) |

# Recap Bonds - intramolecular

| Bond type | How it forms | Key characteristics | Schematic representation | Example |
|---|---|---|---|---|
| **Covalent (nonpolar/polar)** | • Sharing of valence electrons between atoms (orbital overlap) | • Directional<br>• Single, double or triple bonds<br>• Nonpolar or polar depending on ΔEN | **Overlap of atomic orbitals**<br>  <br>Shared electron density between nuclei | **Example: $H_2O$ (polar covalent)**<br>  |
| **Ionic** | • Transfer of electrons between atoms (formation of oppositely charged ions) | • Non-directional<br>• Strong electrostatic attraction<br>• Usually high melting points<br>• Conducts electricity when molten or dissolved | **Electrostatic attraction**<br>  <br>Attraction between oppositely charged ions | **Example: NaCl (ionic)**<br>  |
| **Metallic** | • Attraction between metal cations and delocalised electrons | • Non-directional<br>• Good electrical and thermal conductivity<br>• Malleable and ductile | **"Electron sea" model**<br>  | **Example: Cu (metallic)**<br>  |
| **Coordinate covalent (dative)** | • Both electrons in a shared pair are donated by one atom (donor) to another atom (acceptor) | • Type of covalent bond<br>• Often involves lone pairs<br>• Common in metal complexes | **Lone pair donation**<br>  | **Example: $[Cu(NH_3)_3]^{2+}$ (coordinate bonds)**<br>  |

## How bonding determines material properties

| Property | Ionic solids | Metallic solids | Molecular covalent solids |
|---|---|---|---|
| **Electrical conductivity (solid)** | No | Yes | Usually no |
| **Electrical conductivity (molten or dissolved)** | Yes (molten or in solution) | Yes | Usually no |
| **Melting point** | Generally high | Variable, often high | Generally low (except network covalent) |
| **Mechanical behaviour** | Hard, brittle | Malleable, ductile | Often soft |
| **Typical examples** | NaCl, MgO | Cu, Al, Fe | Ice ($H_2O$), $CO_2$ (dry ice), $I_2$ |

# Types of bonds (intermolecular)

# Hydrogen bond

”Hybrid” covalent and electrostatic. Can be both inter and intramolecular!

H acts as a “bridge” between an atom it’s covalently bound to, and another electronegative atom

Often happens with N, O and F



Hydrogen bonds leads to:

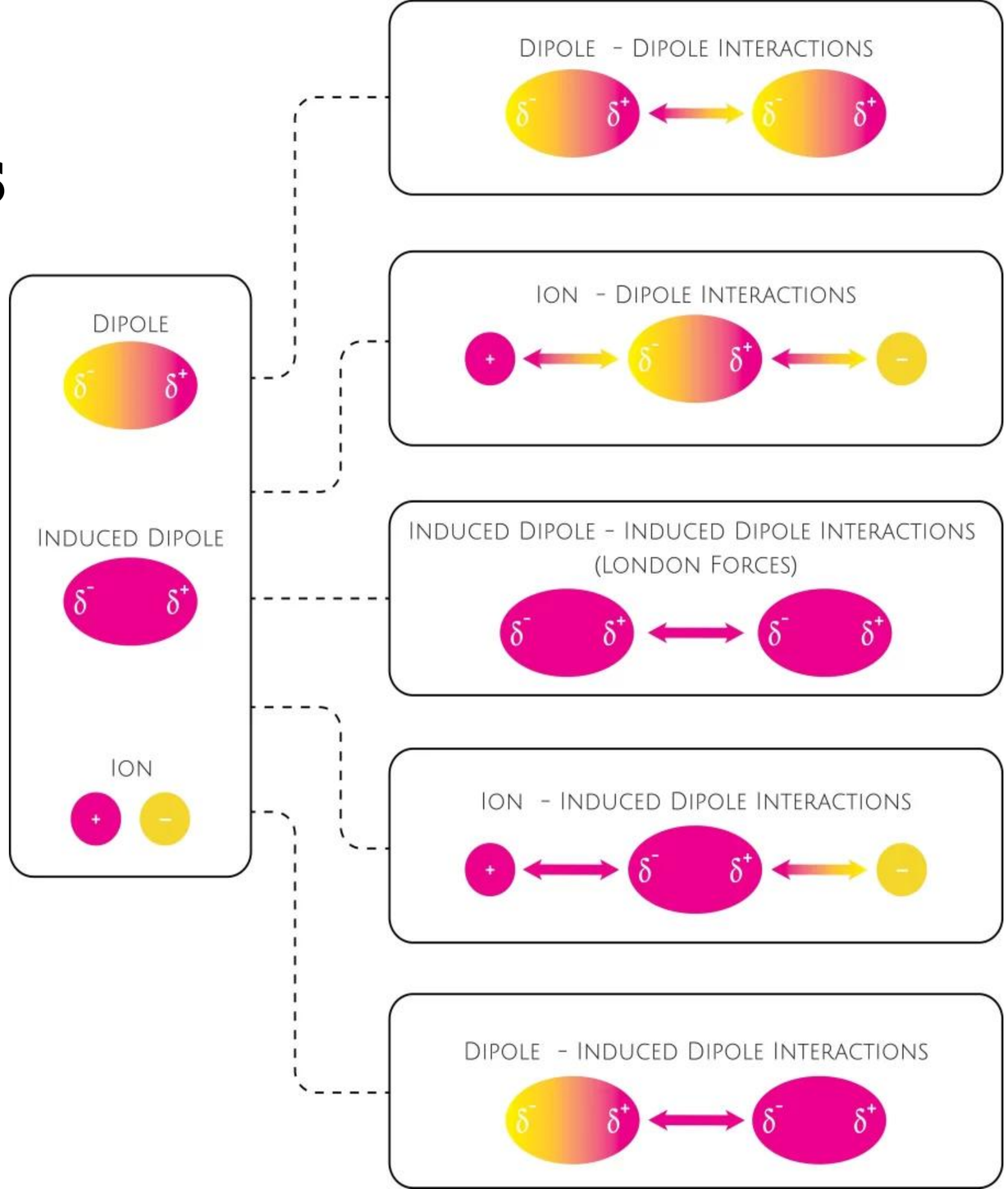
- High viscosities, high surface tension
- Higher melting/boiling points
- Solubility in polar solvents

# Van Der Waals forces

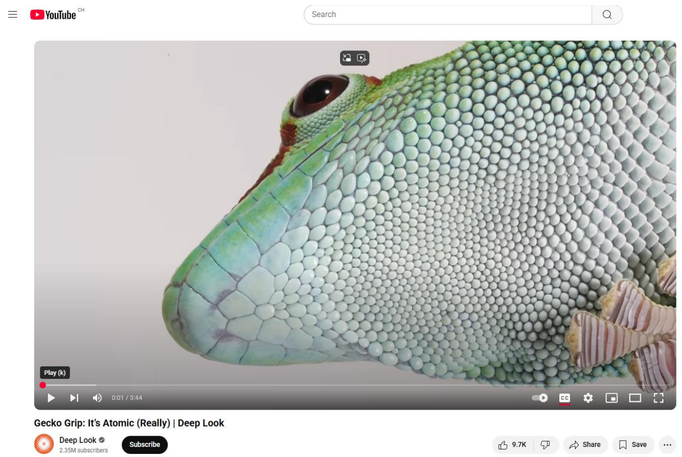
- Family of temporary, distance-dependent electrostatic interactions. Vanish at longer distances
- Generally considered rather weak.
- Caused by temporary imbalances in $e^-$ that create temporary or permanent dipoles.
- Compounds that are more polarizable (more $e^-$) give rise to stronger VdW

VdW forces lead to:

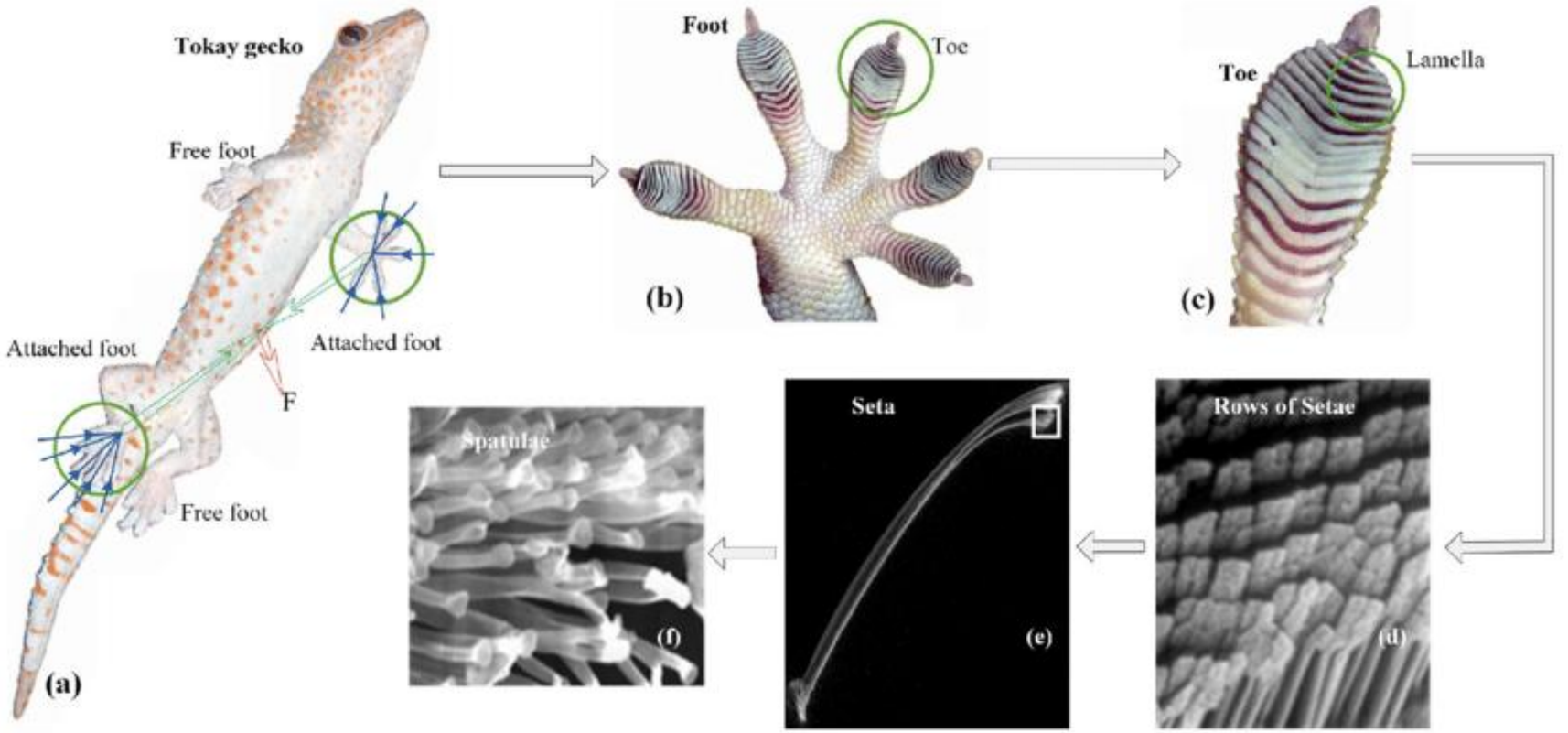
- Generally low melting points



# Van Der Waals – The Gecko case study




https://youtu.be/r5cqJTU5D5I?si=CkfZc0rsO3neqcMH




DOI: 10.1007/s42235-021-00088-7

# Exercise



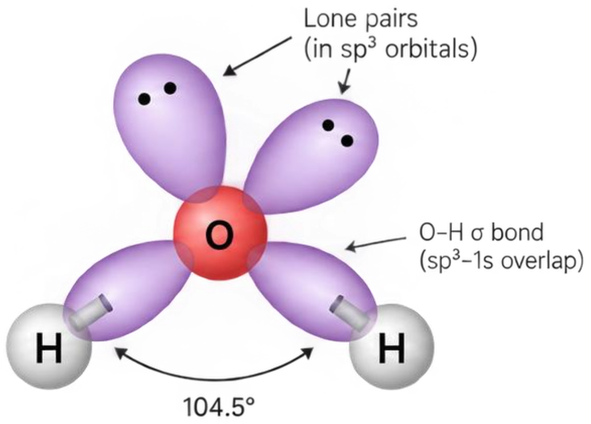
# Break
# 09:10 – 09:25

# Lewis formulas

# Lewis formula for bonds

- Drawing molecules through hybridisation is quite complex!
- Especially with atoms with d, f orbitals…
- There’s a simpler method!

**Hybridisation model (detailed 3D representation)**

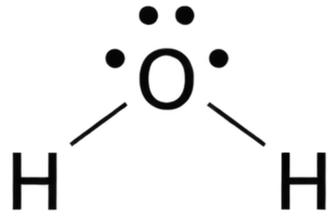



- Shows which orbitals are involved ($sp^3$ on O, 1s on H)
- Shows the 3D shape of the orbitals and the molecular geometry
- Explains the bond angle (104.5°) and arrangement due to electron pair repulsion
- More detailed, but also more complex

**More detailed, shows the 3D electron arrangement and explains the molecular shape.**

**Same molecule ($H_2O$)**

**Lewis structure (simplified 2D representation)**




$H_2O$

- Shows which atoms are bonded (connectivity)
- Shows valence electrons (bonding pairs and lone pairs)
- 2D representation
- Does not show the 3D shape or the orbitals

**Simple and useful for connectivity and electron counting.**

# Lewis formula for bonds

Lewis or electron dot formula is a way of representing the bonding between atoms of a molecule, as well as the lone pairs of electrons that may exist.

Rules :

1. Count the number of valence $e^-$
2. Place the least electronegative atom in the centre of the molecule (excluding hydrogen)
3. Distribute single bonds between all atoms. Each pair of connected $e^-$ in the molecule is represented by a line
4. Count the $e^-$: remaining $e^-$ are represented as a dot → 1 dot = 1 $e^-$
5. Unpaired $e^-$ are shared to form an additional bond and satisfy the octet rule
6. Atoms with higher electronegativity fill the valence shell first
7. Now check if all atoms reach the octet. If not, assign a charge

Exceptions to the rule:
https://chem.libretexts.org/Bookshelves/General_Chemistry/Book%3A_General_Chemistry%3A_Principles_Patterns_and_Applications_(Averill)/08%3A_Ionic_versus_Covalent_Bonding/8.06%3A_Exceptions_to_the_Octet_Rule

# Exercise

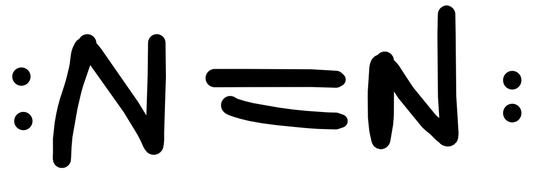
$CH_4$

# Exercise

HCl

# Exercise

Is this correct?



# Lewis formula for bonds – formal charges

After the general structure has been drafted, then formal charges need to be checked and assigned!

Rules :

1. Count the valence electrons of each atom (from the periodic table).
2. Calculate the formal charge for each atom using:

$$\text{Formal charge} = (\text{valence } e^-) - (\text{nonbonding } e^-) - \frac{1}{2}(\text{bonding } e^-)$$

3. Sum of all formal charges = total charge of the molecule or ion.
4. The most stable structure has the smallest possible formal charges.
5. Negative charges should be on the most electronegative atoms.
6. Zero formal charge is preferred whenever possible

# Exercise

$NO_3^-$

# Valence shell electron pair repulsion (VSEPR)

Valence $e^-$ repel each other → the strongest the repulsion, the more unstable the molecule!

→VSEPR predicts geometries of molecules to minimize repulsion

**"AXE" notation**:

A = Central atom (usually least electronegative or atom that forms most bonds)

X = Number of atoms bound to central one

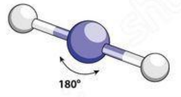
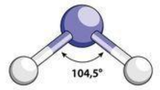
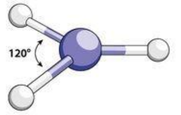
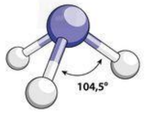
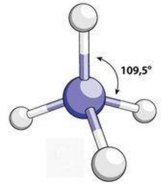
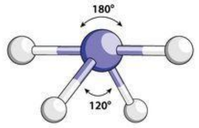
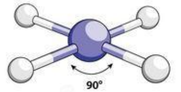
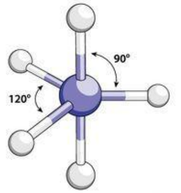
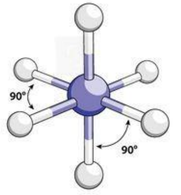
E = Number of free $e^-$ pairs

# Bond length and angles

Generally, atoms arrange themselves far from free $e^-$ in precise geometrical shapes.

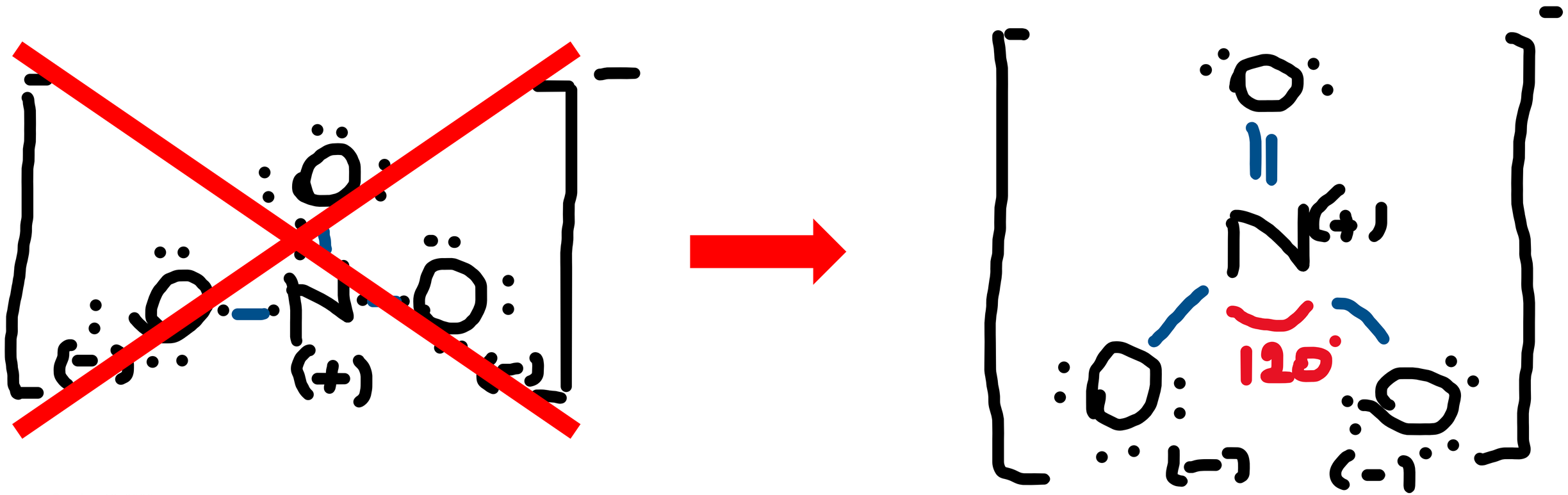
3D representation of AXE configuration:

https://chem.libretexts.org/Bookshelves/Inorganic_Chemistry/Supplemental_Modules_and_Websites_(Inorganic_Chemistry)/Molecular_Geometry/Molecular_Geometry_Overview

| Molecular Formula | VSEPR Notation | Molecular Geometry | Geometric Shape | Example |
|---|---|---|---|---|
| $AX_2$ | $AX_2$ |   | Linear | $CO_2$ |
| | $AX_2E_2$ |   | Bent(V-shaped) | $H_2O$, $OF_2$ |
| $AX_3$ | $AX_3$ |   | Trigonal Planar | $BH_3$, $SO_3$ |
| | $AX_3E$ |   | Trigonal Pyramidal | $NH_3$, $PCl_3$ |
| $AX_4$ | $AX_4$ |   | Tetrahedral | $CH_4$ |
| | $AX_4E$ |   | Seesaw | $SF_4$ |
| | $AX_4E_2$ |   | Square Planar | $XeF_4$ |
| $AX_5$ | $AX_5$ |   | Trigonal Bipyramidal | $PCl_5$ |
| $AX_6$ | $AX_6$ |   | Octahedral | $SF_6$ |

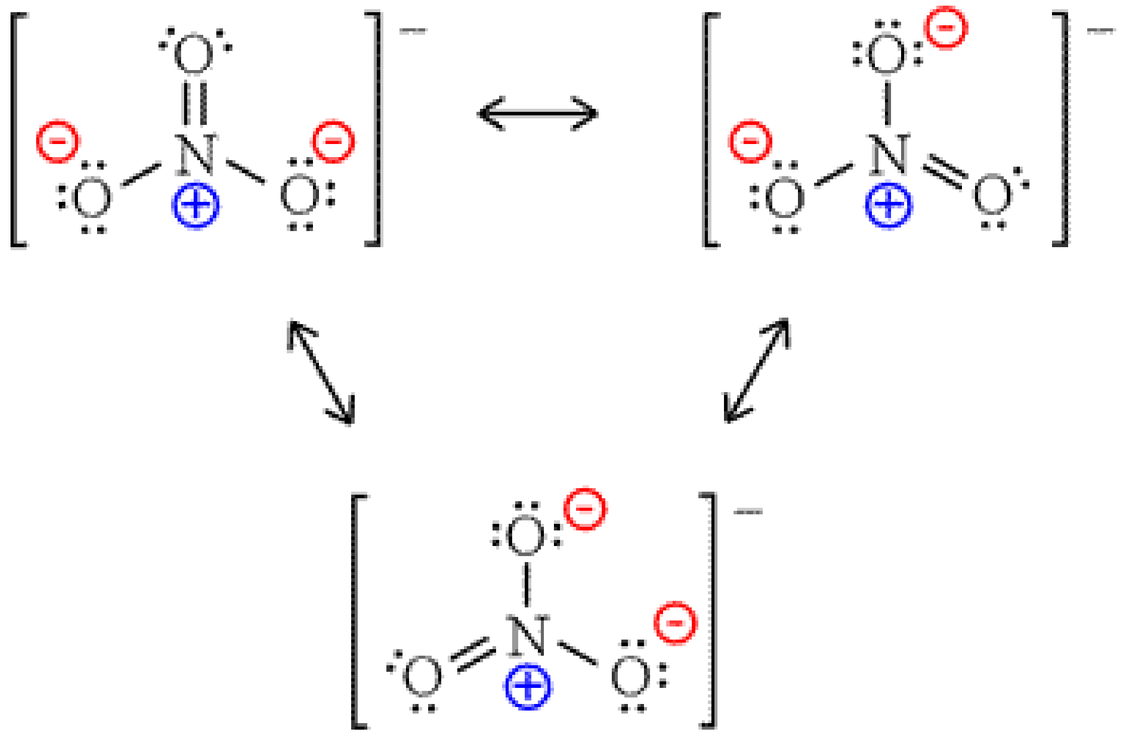
# VSEPR

$NO_3^-$



# Resonance

$NO_3^-$



# Exercise

$SO_4^{2-}$

# Exercise

$SO_4^{2-}$

- Elements in group 3 and below can have “expanded” octet!
  → up to 12 e- !
- Only non-metal elements

# Exercise – Lewis with VSEPR

HCN

# Individual Exercise – Lewis with VSEPR

$PF_3$

# Break
# 10:10 – 10:30

# Individual Exercise – Lewis with VSEPR

$SO_2$

# Individual Exercise – Lewis with VSEPR

$H_2S$

# Discussion + Q&A





# Outlook for Next Week:

- Chemical reactions
- Chemical equilibria
- Concepts of spontaneity
- Entropy, enthalpy, Gibb’s free energy